USN: [23MADIP01] 19/12/24 MIS

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024
FOUNDATION OF MATHEMATICS - 1

Time: 3 Hours Maximum Marks: 50

Sl.No	Questions	Option - A	Option - B	Option - C	Option - D	Answer
1	If y=u±vare the functions of x then dy/dx=.....	dy/dx=du/dx+dv/dx	dy/dx=du/dx∓dv/dx	dy/dx=du/dx±dv/dx	dy/dx=du/dx-dv/dx	
2	If y=uv are the functions of x then dy/dx=....	dy/dx=du/dx dv/dx	dy/dx=u dv/dx+v du/dx	dy/dx=u dv/dx-v du/dx	dy/dx=v du/dx-u dv/dx	
3	If y=u/vare the functions of x then dy/dx=.....	dy/dx=(v du/dx+u dv/dx)/u^2	dy/dx=(v du/dx-u dv/dx)/u^2	dy/dx=(v du/dx+u dv/dx)/v^2	dy/dx=(v du/dx-u dv/dx)/v^2	
4	If y=x^n then dy/dx=.....	x^n	x^(n-1)	nx^n	nx^(n-1)	
5	If y=a^x then dy/dx=....	a^x loga	a^x	loga	None of these	
6	If y=logx then dy/dx=.....	x	1/x	x^2	1/x^2	
7	If y=e^x then dy/dx=.....	e^x	1/e^x	x	None of these	
8	If y=sinx then dy/dx=.....	sinx	cosx	-cosx	- Sin x	
9	If y=cosx then dy/dx=.....	tanx	-sin (x)	-cos(x)	cot(x)	
10	If y=tanx then dy/dx=.....	cotx	-cotx	(sec)^2(x)	None of these	
11	If u = xy+z^2 where x=t, y=t^2, z=t^3 then find du/dt	3t^2+6t^5	3t^2+6t	3t+6t^5	3t^2+t^5	
12	If u = xy^2+x^2y where x = at, y = 2at then find du/dt	18a^3t^2	a^3t^2	6at	6a^2t^2	
13	If u = xy+z^2 where x=t,y=t^2, z=t^3 then find du/dt at t=2	202	204	201	203	
14	If u = x^3 +y^3 where x = t^2 and y = t^3 then find du/dt	6t^5+9t^8	t^5+t^8	6t+9t^2	None of these	
15	If u = x^3 +y^3 where x = t^2 and y = t^3 then find du/dt at t = 1	10	12	14	15	
16	Jacobian of u and v w.r.t x and y is given by	J((u,v)/(x,y))	J((x,y)/(u,v))	J((u,v)/(u,y))	None of these	

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I.No	Questions	Option - A	Option - B	Option - C	Option - D	Answer
17	In Jacobian, JJ'=	0	1	2	3	
18	Find jacobian of u, v, w w.r.t x, y, z given that u = x + y +z, v = y + z and w = z	0	1	2	3	
19	Find jacobian of u, v, w w.r.t x, y, z given that u = xy^2, v = yz^2 and w = zx^2	9x^2y^2z^2	x^2y^2z^2	xyz	None of these	
20	Find jacobian of u, v, w w.r.t x, y, z given that u = xy^2, v = yz^2 and w = zx^2 at (1, 1, 1)	3	5	7	9	
21	∫ x dx from 1 to 2 =	1	2	(3/2)	None of these	
22	∫ sin (x) dx from 0 to π/2 =	0	1	2	3	
23	∫ (x^2) dx from 0 to 1 =	(1/3)	2	0	1	
24	∫ (e^x) dx from 0 to 1 =	e - 1	e	e + 1	None of these	
25	∫ (cos x) dx from 0 to π/2 =	0	1	2	3	
26	Beta Function is defined as	∫x^m (1-x)^n dx, from -2 to 2 (m,n<0)	∫x^(m-1) (1-x)^(n-1) dx, from 0 to 1 (m,n>0)	∫x^(m-3) (1-x)^2n dx, from -2 to 2 ((m,n=0)	None of these	
27	Gamma Function is defined as	∫e^(-x) x^(n-1) dx, from 0 to ∞ (n>0)	∫e^(-3x) x^(n-2) dx,(n>0) from 0 to ∞	∫e^(-xy) x^n dx, from 0 to ∞ (n>5)	None of these	
28	β(m, n) =	β(m, n)	β(n, m)	β(m, m)	β(n, n)	
29	Γ(n+1) =	nΓ(n)	Γ(n)	Γ(n+1)	None of these	
30	Γ(n+1) =…, if n is a positive integer	n!	Γ(n)	Γ(n+1)	Γ(n-1)	
31	Order and degree of((d^3 y)/(dx^3))^4-6x^2 (dy/dx)^8+e^y = sinx is….	Order=3, Degree=4	Order=4,Degree=4	Order=8, Degree=4	Order=2, Degree=6	
32	Solution of dy/dx+y/x=x^2 is ……	xy=1/4 x^4	xy=1/4 x^4+c	xy=1/4 x^2+c	None of these	
33	If M and N are both homogeneous function of x, y of some degree in the differential equation Mdx+Ndy=0 and ∂M/∂y≠∂N/∂x then Integrating factor is….	1/(Mx+Ny)	1/(Nx+My)	M/(Mx+Ny)	None of these	

Sl.No	Questions	Option - A	Option - B	Option - C	Option - D	Answer
34	∂M/∂y & ∂N/∂x for the differential equation ydx+(x+y)dy=0 is…….	y & x	y & (x+y)	1 & 1	None of these	
35	Solution of y dx + x dy = 0 is	xy = c	x = c	y = c	x/y = c	
36	Is (2x+y+1) dx + (x+2y+1)dy = 0 an exact equation?	Exact	Not exact	Not sure	None of these	
37	Is (x+3y-4) dx +(3x+9y-2) dy = 0 an exact equation?	Exact	Not exact	Not sure	None of these	
38	Is (x^2+y^2+x) dx + (xy) dy = 0 an exact equation?	Exact	Not exact	Not sure	None of these	
39	e ^ [∫Pdx] and e ^ [∫Pdy] are called	Multiplicative Factor	Factor	Integrating Factor	None of these	
40	An equation which involves atleast one derivative of y with respect to x is called	Exact equation	Complete equation	Ordinary differential equation	Partial differential equation	
41	The general solution is of the form y=yc+yp, where yp is called ……..	Auxiliary equation	Particular integral	Complimentary function	None of these	
42	The general solution of a non-homogenous equation consists of …….. parts.	1	3	4	2	
43	The general solution of a homogenous equation consists of …….. parts.	1	3	4	2	
44	Differential equation is said to be linear if the dependent variable and its derivatives are of	First degree	Second degree	Third degree	None of these	
45	An equation of the form a0 (ax+b)^2 y"+a1 (ax+b)y'+a2 y= φ(x) is called …….	Cauchy's equation	Legendre's equation	Linear equation	None of these.	
46	An equation of the form a0 x^2 y"+a1 xy'+a2 y= φ(x) is called …….	Linear equation	Legendre's equation	Cauchy's equation	None of these.	
47	Cauchy's linear equation is a particular case of Legendre's linear equation whena=.... and b=...	2 and 3	1 and 3	1 and 2	1 and 0	
48	The operator 1/Dthat stands for the integral is called an …….	Differential operator	Inverse Differential operator	Constant	None of these	
49	To convert Legendre's equation a0 (ax+b)^2 y"+a1 (ax+b) y'+a2 y = φ(x) to linear differential equation of constatnt coefficients, the substitution ax+b=.....	e^t	e^(-t)	t	None of these	
50	To convert Cauchy's equation a0 x^2 y"+a1 xy'+a2 y=φ(x) to linear differential equation of constatnt coefficients, the substitution x.....	e^t	e^(-t)	t	None of these	